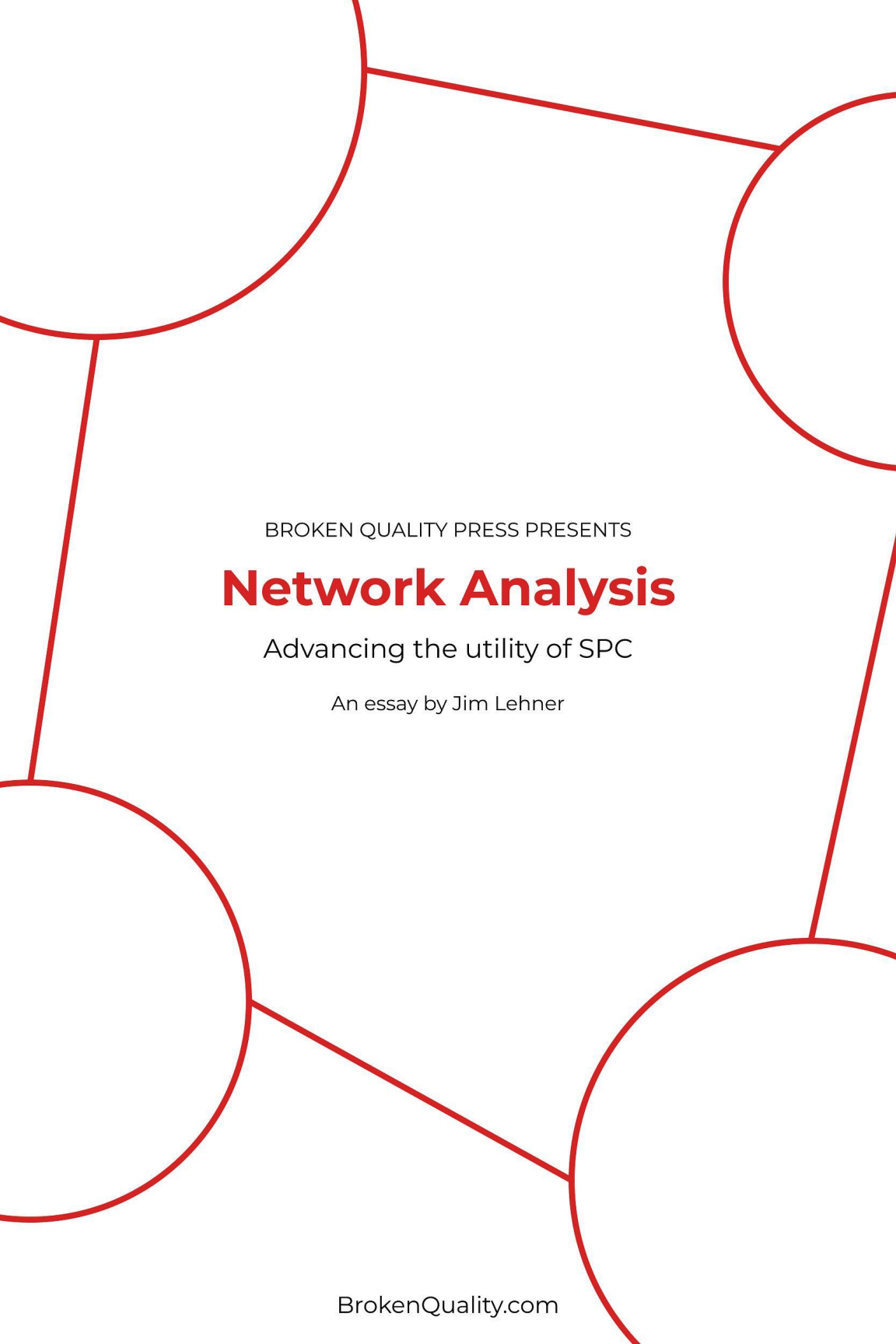


The background of the cover features a minimalist, abstract design composed of several large, thin red circles and straight red lines of varying lengths. These elements are interconnected, creating a network-like structure that frames the central text. The design is clean and modern, with a strong emphasis on the color red.

BROKEN QUALITY PRESS PRESENTS

Network Analysis

Advancing the utility of SPC

An essay by Jim Lehner

JIM LEHNER

Network Analysis

Advancing the utility of SPC

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First edition

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Network Analysis

“While we were busy arguing about how good the parts had to be, they were working hard at making all the parts exactly the same.”¹

—John Betti of Ford Motor Company

An evolutionary inflection point

While Statistical Process Control (SPC) and process behavior charts² have not seen the widespread adoption that they deserve, for those who are familiar with the method and tool, the value is understood. Unlike fishbone diagrams, Pareto charts, and 5S, process behavior charts are not just another tool in the improvement toolkit. Regardless of industry or application, they are a sovereign methodology distinct from the body of work known as statistics. They allow individuals, teams, and organizations to characterize past process behavior so they can anticipate the future. They ensure that the time and attention of technical resources are directed to where they are needed most when they are needed most.

Given these accolades, it is easy to assume that process behavior charts may have reached their evolutionary zenith. After all, since their creation by Dr. Walter Shewhart at Bell Labs in the mid-1920s, process behavior charts have been exposed to more than a century of practical use and refinement. And

¹ *Advanced Topics in Statistical Process Control*, Donald J. Wheeler, (SPC Press, 1995), p. 197.

² The term process behavior chart is synonymous with control chart.

yet, as it stands today, the typical application of process behavior charts is not markedly different from that of the 1920s. As it stands today, the aim of analysis using process behavior charts tends toward the myopic. It tends to focus on understanding the variation associated with singular processes as opposed to networks of elements and the behavior of systems.

While the typical approach to understanding variation using process behavior charts has proven itself time and time again, our capacity to collect and manipulate data at ever increasing volumes and scales has exposed an evolutionary inflection point. Why continue to focus on only individual elements of a system when we can perform analysis that includes many? Why take only a myopic view when we can perform an analysis that is holistic?

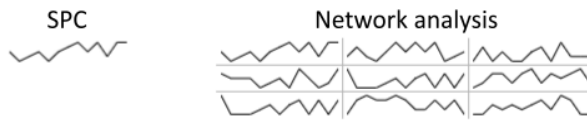


Fig 1: Network analysis expands the logic of SPC using a grid of small multiples. This facilitates direct visual comparison of the elements in the grid and promotes uninterrupted visual reasoning.

Network analysis is a response to these questions and the inherent deficiencies associated with the sequential analysis of individual processes. By collocating the process behavior charts for multiple system elements on a grid that populates a single field of view, an integrated perspective of process behavior is revealed. When the elements of the grid display the same quality characteristic, network analysis enables direct visual comparison between the means and process limits. When the elements of the grid display different quality characteristics but are related in some way—they are quality characteristics from the same part for example—network analysis provides an overview of which characteristics are behaving predictably and which are behaving unpredictably. It highlights which grid elements require attention now and which can wait until later.

Regardless of the nuance in application, network analysis expands our capacity to understand variation. It liberates process behavior charts from the myopic assumption that sequential analysis of singular processes is the only way we can facilitate improvement. It is how we bring process behavior charts into the 21st century and more overtly align the method with Dr. W. Edwards Deming's definition of a system.

Deming, a contemporary and friend of Dr. Shewhart, defined a system as *"a network of interdependent components that work together to try to accomplish the aim of the system."*³ Although the typical application of process behavior charts to singular processes serves as a path for actualizing this definition, until recently, the volume and scale of data that could be leveraged in these applications was limited. With these constraints lifted thanks to the widespread adoption of Industry 4.0 and the Internet of Things (Iot), it is due time that the analytical capacity of process behavior charts was expanded to reflect modern business and manufacturing ecosystems.

Network analysis is the mechanism and approach by which this is achieved. It is how we enable a holistic view of process behavior that better directs the time and attention of technical resources to where they are needed most when they are needed most. It is how we align the use of process behavior charts with the data driven realities of manufacturing and business in the 21st century.

The typical approach

The typical approach to process behavior charts is defined by the sequential analysis of singular processes. The manufacturing process in Figure 2 serves as an example.

Composed of eight machines producing the same part, this process was assumed to be operating predictably. However, upon the discovery of non-conforming product in the product stream, a series of foreseeable, necessary, and costly actions ensue. Most notably, management demands a root cause

³ *The New Economics for Industry, Government, and Education*, Dr. W. Edwards Deming (MIT Press, 1993), p. 25.

be identified, corrective actions be taken, and control plans be put in place.

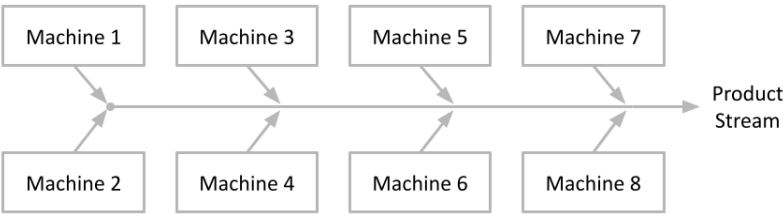


Fig 2: The schematic for a manufacturing process composed of eight machines that produce the same part.

In response to these demands, the manufacturing team gets to work. After reviewing lot numbers and date codes, the non-conforming product is traced to Machine 5. Subsequently, a process behavior chart for this machine is generated. Shown in Figure 3, the process behavior chart for Machine 5 characterizes the process as predictable. This means that, moving forward, the team can expect Machine 5 to produce parts with lengths as long as the Upper Process Limit and as short as the Lower Process Limit without cause for concern. But if this is the case, why then did Machine 8 produce non-conforming product in the first place?

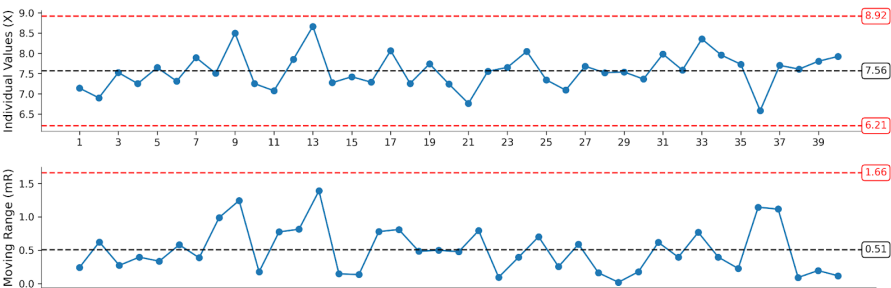


Fig 3: With all the values falling inside the process limits, the process behavior chart for Machine 5 characterizes the process as predictable.

The answer to this question is revealed by the capability histogram in Figure 4. Although the process behavior chart for Machine 5 characterized this process as predictable, the process is biased toward the Upper Specification Limit. This bias, in conjunction with the predictable characterization, indicates that this process will reliably and repeatedly produce non-conforming product now and into the future. This insight is reaffirmed by the differences in the Capability Indices (C_p and C_{pk}) and the Performance Indices (P_p and P_{pk}). As the delta between the target and mean converge, so too do the values for the Capability Indices and the Performance Indices. As the delta between the target and mean diverge, so too do the values for the Capability Indices and the Performance Indices.

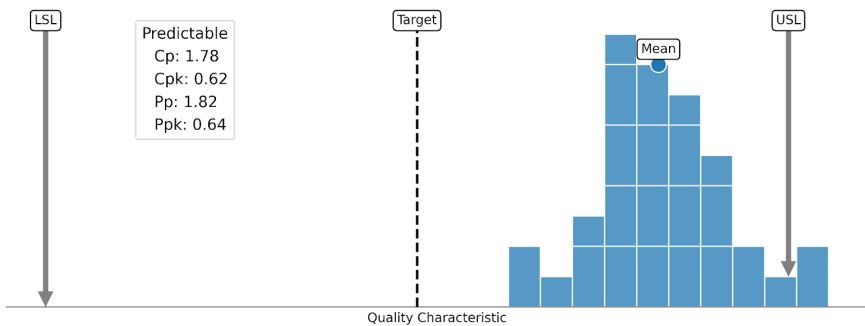


Fig 4: The capability histogram of Machine 5 reveals that while the process was characterized as predictable it is not operating on-target.

In order to stop non-conforming material from being produced in the future, Machine 5 must be centered. Centering is the task of adjusting a process such that the delta between the mean and the target is minimized. It is only through such an effort that the team can expect Machine 5 to reliably and repeatedly produce conforming product in the future.

After making the requisite adjustments to better align the process mean with the target, a second process behavior chart and capability histogram for Machine 5 were generated. The combined X Chart and capability histogram are

shown in Figure 5. Prior to the centering effort the process was characterized as predictable but not performing on-target, after the centering effort the process is still characterized as predictable but now operates very near the target.

As to be expected, these improvements are reflected in the associated process capability indices. With the delta between the target and mean now smaller, the Capability Indices (C_p and C_{pk}) and the Performance Indices (P_p and P_{pk}), while not exactly the same (they never will be), converge. While Machine 5 should be monitored for drift and future process improvements should aim to further reduce the variation, for all intentions and purposes the problem of Machine 5 producing non-conforming product has been addressed.

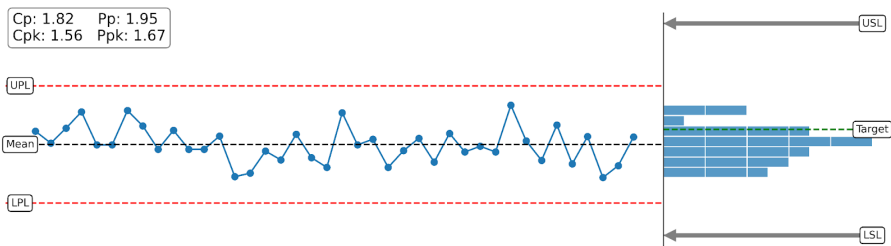


Fig 5: The X Chart and capability histogram for Machine 5 after centering.

The prior discussion is a quintessential example of the unexpected production of non-conforming product initiating action and a myopic set of improvement efforts ensuing as a result. While the manufacturing team was certainly successful in their efforts to identify and eliminate the root cause of the non-conforming product, their focus on Machine 5 leaves the seven other machines producing the same product operating in a state of business as usual. What insights does this approach leave hidden and what opportunities does it overlook? Are other machines that are operating near the specification limits poised to produce non-conforming product? If so, how much should these machines be adjusted? Can the same adjustments that were made with Machine 5 be made with other machines?

To answer these questions using the typical approach, process behavior charts for each machine must be generated and the requisite actions based on their insights taken. Although such efforts are by no means an impossible task, their execution imparts no formal sequence. Based on their current view, the manufacturing team can start with Machine 1 or start with Machine 8. For all they know either action is the same. The team lacks the necessary information to make data driven decisions and establish a plan about where to begin. Luckily, there is an alternative to this haphazard approach. Rather than rely on gut, guess, or additional loss to initiate actions, network analysis facilitates a holistic view. This view reveals which machines need immediate attention and which machines can wait until later. It reveals which, if any, machines are operating *on-target with minimum variance* and which require modifications.

Network analysis

The network analysis for the manufacturing process from above, after the adjustments were made to Machine 5, is shown in Figure 6. This integrated perspective of all eight machines is enabled by a shared y-axis that allows for direct visual comparisons of means and process limits. This makes several otherwise overlooked insights clear.

First, machines 1 through 7 are characterized as predictable. This indicates that moving forward the manufacturing team can expect product from each of these machines to fall within their respective process limits. Such insights are invaluable given that it appears some machines, like Machine 4, require centering similar to that performed on Machine 5. In the absence of a predictable characterization, centering efforts are a waste of time. When a process is characterized as unpredictable, numerical summaries like the mean are little more than historical descriptions. While they may represent what has happened in the past this cannot be generalized to the future in any meaningful way. Consequently, making adjustments based on the mean of an unpredictable process is likely to make things worse more than it is likely to make things better.

Second, with one value greater than the Upper Process Limit, Machine 8 is characterized as unpredictable. This is a signal that assignable causes of exceptional variation are influencing Machine 8's behavior. Although at the moment all product produced on Machine 8 is conforming, until the assignable causes of exceptional variation are eliminated the potential for non-conforming product will persist into the future.

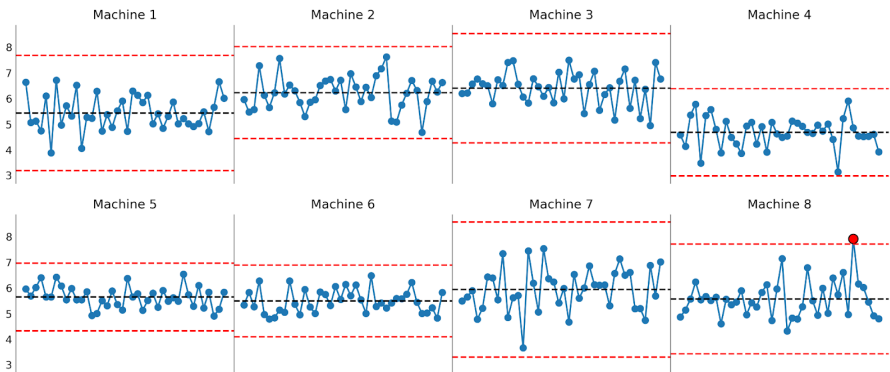


Fig 6: Network analysis for a manufacturing process composed of eight lines all producing the same part.

Third, the network analysis reveals that the magnitude of the variation influencing the behavior of each machine is unique. Where the variation associated with some machines is comparatively large, machines 3 and 7 for example, other machines display variation that is comparatively small, like machines 5 and 6. In either case, the eight machines that constitute this manufacturing process display somewhat unique variation even though they are producing the same part. This presents the manufacturing team with the opportunity to cross-pollinate the manufacturing process with knowledge. Knowledge gained by examining and understanding the underlying cause of machines that exhibit comparatively less variation can be transferred to machines that exhibit comparatively more. Such efforts reduce the machine-to-machine variation. In doing so, the manufacturing process at large will

produce more uniform product.

The notion of variation between different machines was overlooked by the manufacturing team during their initial analysis. Their myopic focus on Machine 5, while a successful effort, produced a myopic result. It instantiated change only on Machine 5 and, in doing so, limited the impact of the knowledge gained. While oftentimes a myopic focus is necessary to complete the task at hand, we should never forget about the larger objective. We should never forget that our task is not to argue about how good the parts have to be from a single machine. We should be working to make all the parts exactly the same regardless of which machine produced them.

Network analysis limit chart

While by now the benefits of a network analysis based approach to understanding variation and facilitating improvement should be clear, additional texture is revealed when the manufacturing data is plotted in a network analysis limit chart. Such a chart is shown in Figure 7. Where the network analysis in Figure 6 plotted the manufacturing data in the context of the process limits, the network analysis limit chart in Figure 7 plots the data in the context of the specification limits (gray), target (green), and mean (black). Such a view reveals insights that the network analysis from Figure 6 only hinted at or completely obscured.

Perhaps the most notable insight from Figure 7 are the positions of the means relative to the target. Where we inferred that the position of several machines required adjustments based on the location of their means from Figure 6, Figure 7 transforms this inference into evidence. In particular it highlights that adjustments should be made in earnest to Machine 4. This is because, as it currently stands, Machine 4 has recently produced non-conforming product. Given its predictable characterization, the manufacturing team can expect these non-conformances to continue until adjustments are made that better align the mean of Machine 4 with the target. In addition, machines 4, 1, 3, and 6 could also benefit from a centering exercise.

NETWORK ANALYSIS

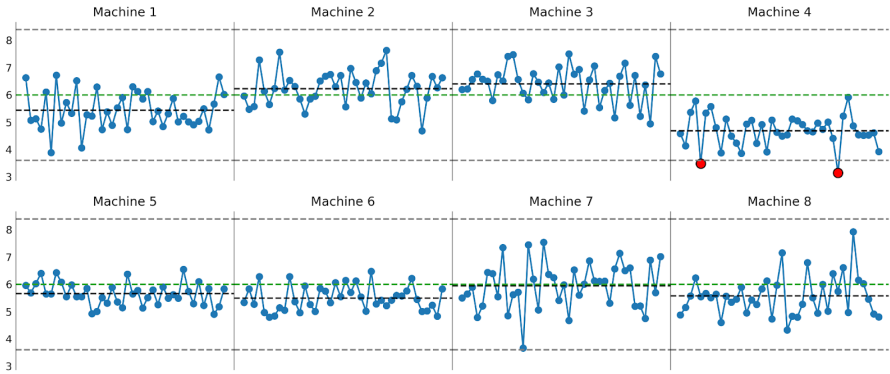


Fig 7: The network analysis limit chart presents the process data in the context of the specification limits (gray), target (green), and mean (black).

While it appears that Machine 8 would also benefit from centering, its behavior has been characterized as unpredictable. Until the assignable causes of exceptional variation that are influencing Machine 8 are understood and eliminated, any efforts to center or adjust this process will be futile. An unpredictable process is described as unpredictable because it behaves unpredictably. While it is possible to calculate summary statistics for an unpredictable process, these statistics will bound and shift in ways that are also unpredictable. This makes it very unlikely that adjustments made using these always changing summary statistics will produce a desired result. Instead, time and attention is better spent understanding and eliminating the assignable causes. When, and only when, an unpredictable process is shown to be predictable, should efforts to center the process be made.

All this discussion of centering raises an important question: *how much should each of the eight machines in this process be adjusted to center them at the target?* By subtracting the target value from the mean of each machine, an approximate centering distance can be calculated.

$$\text{Centering Distance} = \text{mean} - \text{target}$$

While it is unlikely that any machine based features will possess the necessary fidelity to make such fine adjustments, the approximate centering distances, shown as a bar chart in Figure 8, serve as a guide.

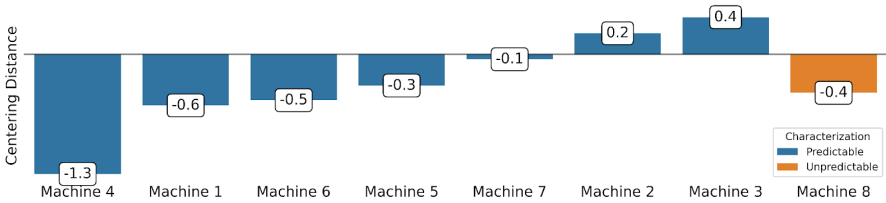


Fig 8: The centering distances for each of the eight machines.

Grouped according to characterization then sorted from smallest negative deviation to largest positive deviation, the bar chart in Figure 8 makes clear where the manufacturing team should begin their centering efforts. As they may have guessed, Machine 4 has the largest deviation of -1.3 followed by machines 1 and 6. The respective deviations for these machines is -0.6 and -0.5. The sign of the centering distances is informative. A negative centering distance indicates that the mean is “below” the target while a positive centering distance indicates that the mean is “above” the target. While this convention may not directly correlate to up and down in physical space, it ensures that the logic used to facilitate adjustments is the same for each centering effort.

In the same way that we can approximate the centering distance for each machine in the manufacturing process, we can approximate the required reductions in variation using the process limit range and tolerance. The process limit range is the distance between the Upper Process Limit and the Lower Process Limit. By subtracting the tolerance from the process limit range for each machine, a value called the tolerance delta can be calculated.

$$Tolerance\ Delta = process\ limit\ range - tolerance$$

When the tolerance delta is negative, the routine variation associated with a particular machine is less than the tolerance. When the tolerance delta is positive, the routine variation associated with a particular machine is greater than the tolerance.

As was the case with the centering distances, displaying the tolerance deltas in the form of a bar chart, like that shown in Figure 9, is enlightening. Grouped by characterization then sorted from largest positive tolerance delta to smallest negative tolerance delta, the bar chart in Figure 9 reveals where reductions in variation would be most beneficial. In particular it highlights that the variation associated with Machine 7, with a positive tolerance delta of 0.5, is greater than the specified tolerance. Given that the behavior of this machine was characterized as predictable and it is operating nearly on target, it is a distinct possibility that Machine 7 will at some point in the future produce non-conforming product. Moving forward, the manufacturing team must make efforts to ensure that, at the very least, the tolerance delta for Machine 7 is negative. How can they do that? The answer lies with the other machines.

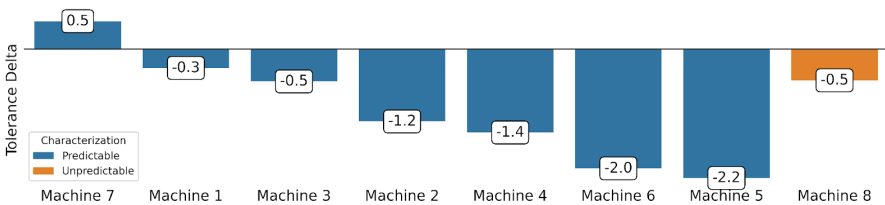


Fig 9: The tolerance deltas for each of the eight machines.

In the same way that the tolerance deltas in Figure 9 highlight Machine 7 as requiring a reduction in variation, machines 5 and 6 are also highlighted. The tolerance deltas for these machines are of -2.2 and -2.0 respectively. Compared to the other machines this is relatively minimal variation. As discussed earlier, when a tolerance delta is negative, the routine variation associated with a process is less than the tolerance. The negative tolerance deltas for machines 5 and 6 are proof that these machines are capable of

operating with reduced variation.

As the manufacturing team moves forward, it would be worth their time to investigate the process settings and procedures associated with machines 5 and 6 and compare those process settings with the procedures associated with machines the other machines. Such efforts are likely to reveal differences that, when applied to machines 7, 1, and 3, reduce the variation. Given that our aim should always be to make parts as similar as possible regardless of the machine that produces them, uniform variation from all eight machines would be a comprehensive step forward.

As was the case with the centering process, Machine 8 is omitted from efforts to reduce common causes of routine variation. This is because, at present, it is still influenced by assignable causes of exceptional variation. After the assignable causes have been eliminated, the same logic applied to the other machines to center the processes and reduce variation should be applied to Machine 8.

Different methods produce different results

In contrast to the typical application of process behavior charts, network analysis expands the scope and scale of insights available to improvement efforts and teams. Where the typical approach is defined by a myopic focus on a singular process or specific element of a system, network analysis is defined by a holistic view of many. Where the typical approach reveals localized insights, network analysis reveals insights that can be generalized. Naturally, these different methods produce different results and these different results have varying impacts.

Where the typical approach is an effective response to a specific problem, such a response has a narrow impact. It ensures that the insights uncovered and knowledge gained are only applied to the process element under consideration. While this is certain to stop bad product from being produced by a specific element of a system, it does nothing to stop bad product from being produced by other elements of the same system.

Remember, Deming defined a system as *“a network of interdependent*

components that work together to try to accomplish the aim of the system."⁴ To evaluate and understand process behavior of a single component even though there are many components performing the same task is to reject Deming's definition. It is to reject the reality that non-conforming product has more than one avenue by which it can be produced. Without network analysis, luck plays an understated role in the success of improvement efforts. But why rely on luck when we can rely on analysis? Why take actions only in response to loss when investigation and planning can be used to avoid problems altogether?

Typical approach	Network analysis
Sequential analysis	Parallel analysis
Myopic & singular	Holistic & many
Gut, guess, loss	Analysis, insight, planning
Localized insights	Generalized knowledge

Fig 10: Differences between the typical approach and network analysis.

Although there will always be unknown unknowns, network analysis uncovers more of these unknown unknowns than the typical approach and application to SPC. It trades the chaos of gut, guess, and loss, for analysis, planning, and insight. In doing so, network analysis directs the time and attention of technical resources within a broader improvement framework. This broader framework ensures that improvement is expressed in a continuous cycle rather than a one off task that needs completion. It ensures that we strive to make all parts exactly the same regardless of the machine that produced them rather than argue about how good parts have to be for a customer to buy them.

⁴ *The New Economics for Industry, Government, and Education*, Dr. W. Edwards Deming (MIT Press, 1993), p. 25.

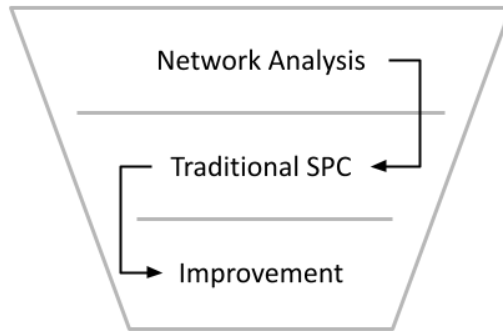


Fig 11: The improvement funnel begins with network analysis and ends with specific improvement efforts.

While it should go without saying, the typical approach and network analysis are not enemies but friends. They are two sides of the same coin that, when used in conjunction with each other, realize improvements that are beyond what is typical. The integrated and holistic view enabled by network analysis should be seen as a funnel. This funnel allows a vast net to be cast across our processes and systems that reveals broad insights. This broad set of insights should be used to establish a plan that will inevitably be expressed by sequential analysis of singular processes i.e. the typical approach to SPC and process behavior charts.

Summary

Today’s distributed networks of business and industrial processes make it increasingly unlikely that sequential analysis of individual processes using process behavior charts will suffice. Rather than reinvent a method that has proven its utility for over one hundred years, it is more productive to explore how the ethos and insights of process behavior charts can be scaled to meet industries current needs.

This is the value and intent of network analysis. Rather than choose where to begin improvement efforts based on gut, guess, or loss, network analysis applies the logic of process behavior charts to a grid of small multiples that

is composed of interdependent components working together to accomplish an aim. The result is a data visualization that is informationally dense and promotes uninterrupted visual reasoning that is unparalleled by any other tool in the improvement space.

Not only does network analysis reveal broad insights in a way that the typical approach to SPC is incapable of doing but, perhaps more importantly, it establishes a holistic plan for improvement and action. Starting with characterization, network analysis reveals which components of a system are behaving predictably and which components of a system are behaving unpredictably. For components characterized as predictable, efforts to center and reduce variation become the priority. This is facilitated with the help of centering distances and tolerance deltas like those displayed in the bar charts above. For components characterized as unpredictable, the tried and true approach applies. For these components assignable causes of exceptional variation must be identified, understood, and eliminated. If, and only if, the influence of assignable causes is eliminated should efforts to center and reduce variation be performed.

Hopefully by now the benefits of a network analysis based approach to process improvement, as opposed to the sequential analysis of singular processes using process behavior charts, is clear. When applicable, network analysis expands the scope and scale of insights that can be used to inform improvement efforts. It pulls what would otherwise be disparate analysis of elements performing the same task under a single roof. This promotes uninterrupted visual reasoning that not only enables a broader set of improvements to be made but, when appropriate, allows knowledge gained from one part of a system to be disseminated to others. Such activities enable rapid learning and rapid improvement that is unrivaled by any other method.

Notes

- The typical approach to SPC is myopic in its efforts to improve singular processes
- Network analysis establishes a holistic view of multiple components

performing the same task

- The shared y-axis of the network analysis grid enables direct visual comparison between the grid elements
- Network analysis promotes uninterrupted visual reasoning
- The typical approach to SPC and network analysis should be used in conjunction to maximize the impact of improvement efforts

Appendix

Figures 6 and 7 of this document were generated using the **network_analysis** and **network_analysis_limit_plot** Python functions. Both functions can be found in the **improvement.py** library. This library is available via [GitHub repository](https://github.com/jimlehner/improvement).

To install **improvement.py** directly from GitHub without cloning the repository, enter the following command using the command prompt:

```
pip install git+https://github.com/jimlehner/improvement
```

After installing the **improvement.py** library, import **process.improvement** as **pi**. The **network_analysis** function can be called via **pi.network_analysis()**. Similarly, the **network_analysis_limit_plot** function can be called via **pi.network_analysis_limit_plot()**.

